

7. Write a program to implement Logistic Regression (LR) algorithm in python

CODE:

```

import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split

def evaluate_dataset(df, feature_cols, target_col, new_sample_dict, title):
    print(f"\n{' '*50}\n{title}\n{' '*50}")

    # Feature Matrix (X) and Target Vector (y)
    X = df[feature_cols]
    y = df[target_col]

    # Split dataset
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.3, random_state=42
    )

    # Train model
    model = LogisticRegression()
    model.fit(X_train, y_train)

    # Predict new sample
    new_sample_df = pd.DataFrame([new_sample_dict])
    prediction = model.predict(new_sample_df)[0]

    print("Training Complete.")
    print("New Sample Data :", new_sample_dict)
    print("Predicted Class :", prediction)

# -----
# Dataset 1: Student Placement Prediction
# -----
d1 = pd.DataFrame(
    {
        "CGPA": [9.2, 8.8, 8.5, 7.2, 6.8, 9.0, 7.0, 8.7, 6.5, 9.1],
        "Aptitude Score": [90, 85, 82, 72, 68, 91, 70, 86, 65, 92],
        "Communication Score": [88, 84, 80, 70, 65, 89, 68, 83, 60, 90],
        "Placement": [1, 1, 1, 0, 0, 1, 0, 1, 0, 1],
    }
)

evaluate_dataset(
    df=d1,
    feature_cols=["CGPA", "Aptitude Score", "Communication Score"],
    target_col="Placement",
    new_sample_dict={
        "CGPA": 8.9,
        "Aptitude Score": 88,
        "Communication Score": 85,
    }
)

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    },
    title="Dataset 1: Student Placement Prediction",
)

# -----
# Dataset 2: Loan Approval Prediction
# -----
d2 = pd.DataFrame(
    {
        "Income": [8, 7, 6, 4, 3, 9, 5, 8, 4, 7],
        "Credit Score": [780, 760, 730, 620, 590, 810, 650, 770, 610, 750],
        "Loan Amount": [5, 4, 5, 8, 9, 4, 7, 5, 8, 6],
        "Loan Approved": [1, 1, 1, 0, 0, 1, 0, 1, 0, 1],
    }
)

evaluate_dataset(
    df=d2,
    feature_cols=["Income", "Credit Score", "Loan Amount"],
    target_col="Loan Approved",
    new_sample_dict={"Income": 7, "Credit Score": 760, "Loan Amount": 5},
    title="Dataset 2: Loan Approval Prediction",
)

# -----
# Dataset 3: Disease Diagnosis
# -----
d3 = pd.DataFrame(
    {
        "Temperature": [
            39.1,
            38.8,
            37.0,
            39.3,
            36.8,
            38.7,
            37.1,
            39.0,
            36.9,
            38.9,
        ],
        "Heart Rate": [112, 109, 78, 115, 74, 110, 80, 113, 76, 111],
        "Oxygen Level": [94, 95, 99, 93, 98, 94, 98, 92, 99, 93],
        "Disease": [1, 1, 0, 1, 0, 1, 0, 1, 0, 1],
    }
)

evaluate_dataset(
    df=d3,
    feature_cols=["Temperature", "Heart Rate", "Oxygen Level"],

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        target_col="Disease",
        new_sample_dict={
            "Temperature": 38.8,
            "Heart Rate": 110,
            "Oxygen Level": 94,
        },
        title="Dataset 3: Disease Diagnosis",
    )

# -----
# Dataset 4: Employee Promotion Prediction
# -----
d4 = pd.DataFrame(
    {
        "Experience": [10, 9, 8, 4, 3, 11, 5, 8, 2, 7],
        "Performance Score": [95, 92, 90, 70, 65, 96, 75, 89, 60, 87],
        "Training Hours": [50, 48, 45, 25, 20, 52, 30, 44, 18, 40],
        "Promotion": [1, 1, 1, 0, 0, 1, 0, 1, 0, 1],
    }
)

evaluate_dataset(
    df=d4,
    feature_cols=["Experience", "Performance Score", "Training Hours"],
    target_col="Promotion",
    new_sample_dict={
        "Experience": 8,
        "Performance Score": 91,
        "Training Hours": 46,
    },
    title="Dataset 4: Employee Promotion Prediction",
)

# -----
# Dataset 5: Customer Purchase Prediction
# -----
d5 = pd.DataFrame(
    {
        "Age": [22, 25, 28, 32, 35, 40, 45, 30, 24, 38],
        "Annual Income": [3, 4, 5, 7, 8, 9, 10, 6, 4, 9],
        "Spending Score": [30, 35, 45, 60, 65, 75, 80, 55, 40, 70],
        "Purchase": [0, 0, 0, 1, 1, 1, 1, 1, 0, 1],
    }
)

evaluate_dataset(
    df=d5,
    feature_cols=["Age", "Annual Income", "Spending Score"],
    target_col="Purchase",
    new_sample_dict={"Age": 34, "Annual Income": 8, "Spending Score": 68},

    title="Dataset 5: Customer Purchase Prediction",
)

```

OUTPUT:

```
===== RESTART: C:/kanishka/MACHINE LEARNING/ML EX 7.py =====  
  
=====   
Dataset 1: Student Placement Prediction  
=====   
Training Complete.  
New Sample Data : {'CGPA': 8.9, 'Aptitude Score': 88, 'Communication Score': 85}  
Predicted Class : 1  
  
=====   
Dataset 2: Loan Approval Prediction  
=====   
Training Complete.  
New Sample Data : {'Income': 7, 'Credit Score': 760, 'Loan Amount': 5}  
Predicted Class : 1  
  
=====   
Dataset 3: Disease Diagnosis  
=====   
Training Complete.  
New Sample Data : {'Temperature': 38.8, 'Heart Rate': 110, 'Oxygen Level': 94}  
Predicted Class : 1  
  
=====   
Dataset 4: Employee Promotion Prediction  
=====   
Training Complete.  
New Sample Data : {'Experience': 8, 'Performance Score': 91, 'Training Hours': 46}  
Predicted Class : 1  
  
=====   
Dataset 5: Customer Purchase Prediction  
=====   
Training Complete.  
New Sample Data : {'Age': 34, 'Annual Income': 8, 'Spending Score': 68}  
Predicted Class : 1  
>>>
```

8. Write a program to implement Linear Regression (LR) algorithm in python

```
import pandas as pd
from sklearn.linear_model import LinearRegression

def run_linear_regression(df, feature_cols, target_col, new_sample_dict, title):
    print(f"=== {title} ===")

    # Feature Matrix (X) and Target Vector (y)
    X = df[feature_cols]
    y = df[target_col]

    # Train model
    model = LinearRegression()
    model.fit(X, y)

    # Predict new sample
    new_sample_df = pd.DataFrame([new_sample_dict])
    prediction = model.predict(new_sample_df)[0]

    print("New Sample Input :", new_sample_dict)
    print(f"Predicted Value : {prediction:.2f}")
    print("-" * 50)

# -----
# Dataset 1: House Price Prediction
# -----
d1 = pd.DataFrame(
    {
        "House Size": [
            800,
            1000,
            1200,
            1400,
            1600,
            1800,
            2000,
            2200,
            2400,
            2600,
        ],
        "Bedrooms": [2, 2, 3, 3, 4, 4, 5, 5, 6, 6],
        "House Age": [10, 8, 6, 5, 4, 3, 2, 1, 1, 0],
        "Price": [35, 45, 55, 65, 75, 85, 95, 105, 115, 125],
    }
)

run_linear_regression(
    df=d1,
    feature_cols=["House Size", "Bedrooms", "House Age"],
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        target_col="Price",
        new_sample_dict={"House Size": 1700, "Bedrooms": 4, "House Age": 3},
        title="Dataset 1: House Price Prediction",
    )

# -----
# Dataset 2: Student Marks Prediction
# -----
d2 = pd.DataFrame(
    {
        "Study Hours": [2, 3, 4, 5, 6, 7, 8, 9, 10, 11],
        "Attendance": [70, 75, 80, 82, 85, 88, 90, 92, 95, 97],
        "Assignment Score": [60, 65, 70, 75, 80, 85, 90, 94, 96, 98],
        "Final Marks": [55, 60, 68, 74, 80, 86, 91, 95, 98, 100],
    }
)

run_linear_regression(
    df=d2,
    feature_cols=["Study Hours", "Attendance", "Assignment Score"],
    target_col="Final Marks",
    new_sample_dict={
        "Study Hours": 7,
        "Attendance": 90,
        "Assignment Score": 88,
    },
    title="Dataset 2: Student Marks Prediction",
)

# -----
# Dataset 3: Employee Salary Prediction
# -----
d3 = pd.DataFrame(
    {
        "Experience": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10],
        "Education Level": [15, 15, 16, 16, 17, 17, 18, 18, 18, 19],
        "Skill Score": [60, 65, 70, 75, 80, 82, 85, 88, 90, 94],
        "Salary": [3.5, 4.2, 5.1, 6.0, 7.2, 8.0, 9.1, 10.0, 11.2, 12.5],
    }
)

run_linear_regression(
    df=d3,
    feature_cols=["Experience", "Education Level", "Skill Score"],
    target_col="Salary",
    new_sample_dict={"Experience": 7, "Education Level": 18, "Skill Score": 86},
    title="Dataset 3: Employee Salary Prediction",
)

# -----

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# Dataset 4: Car Fuel Efficiency Prediction
# -----
d4 = pd.DataFrame(
    {
        "Engine Capacity": [
            1.0,
            1.2,
            1.4,
            1.6,
            1.8,
            2.0,
            2.2,
            2.4,
            2.6,
            2.8,
        ],
        "Vehicle Weight": [
            900,
            950,
            1050,
            1150,
            1250,
            1350,
            1450,
            1550,
            1650,
            1750,
        ],
        "Speed": [60, 70, 80, 90, 100, 110, 120, 130, 140, 150],
        "Mileage": [24, 22, 20, 18, 16, 15, 13, 12, 11, 10],
    }
)

run_linear_regression(
    df=d4,
    feature_cols=["Engine Capacity", "Vehicle Weight", "Speed"],
    target_col="Mileage",
    new_sample_dict={
        "Engine Capacity": 1.5,
        "Vehicle Weight": 1100,
        "Speed": 85,
    },
    title="Dataset 4: Car Fuel Efficiency Prediction",
)

# -----
# Dataset 5: Sales Prediction
# -----
d5 = pd.DataFrame(
    {
        "Advertising Cost": [10, 15, 20, 25, 30, 35, 40, 45, 50, 55],
        "Salespersons": [2, 3, 4, 5, 6, 7, 8, 9, 10, 11],
        "Store Visits": [120, 150, 180, 210, 240, 260, 280, 300, 320, 340],
        "Monthly Sales": [3.5, 4.5, 5.8, 7.0, 8.2, 9.5, 10.8, 12.0, 13.5, 15.0],
    }
)

run_linear_regression(
    df=d5,
    feature_cols=["Advertising Cost", "Salespersons", "Store Visits"],
    target_col="Monthly Sales",
    new_sample_dict={
        "Advertising Cost": 38,
        "Salespersons": 7,
        "Store Visits": 270,
    },
    title="Dataset 5: Sales Prediction",
)

```

OUTPUT:


```
===== RESTART: C:/kanisnka/M
=== Dataset 1: House Price Prediction ===
New Sample Input : {'House Size': 1700, 'Bedrooms': 4, 'House Age': 3}
Predicted Value   : 80.00
-----
=== Dataset 2: Student Marks Prediction ===
New Sample Input : {'Study Hours': 7, 'Attendance': 90, 'Assignment Score': 88}
Predicted Value   : 90.00
-----
=== Dataset 3: Employee Salary Prediction ===
New Sample Input : {'Experience': 7, 'Education Level': 18, 'Skill Score': 86}
Predicted Value   : 9.15
-----
=== Dataset 4: Car Fuel Efficiency Prediction ===
New Sample Input : {'Engine Capacity': 1.5, 'Vehicle Weight': 1100, 'Speed': 85}
Predicted Value   : 18.97
-----
=== Dataset 5: Sales Prediction ===
New Sample Input : {'Advertising Cost': 38, 'Salespersons': 7, 'Store Visits': 270}
Predicted Value   : 10.28
-----
>>> |
```